

UIA Inspector

User Guide

Inspect Windows controls • Build resilient macros • AI-assisted selectors • One-click
code generation

the-Automator.com

Version 2.5.0

Contents

- 01 Overview
- 02 Installation & First Launch
- 03 Quick Start
- 04 Capturing Elements
- 05 The UIA Tree
- 06 Properties & Patterns
- 07 Anchor Points
- 08 Building Macros
- 09 AI Assistance
- 10 Preferences
- 11 Tray Menu
- 12 What's New in 2.5
- 13 Tips & Troubleshooting

01 Overview

UIA Inspector is a stand-alone tool that lets you point at any control in any Windows application — a button, a menu item, a tree node, a tab — and instantly see how to talk to it from AutoHotkey. It then helps you turn those clicks into a clean, working AHK macro you can paste into your own scripts.

Where it really shines is fixing the hardest part of UI automation: *writing selectors that don't break tomorrow*. The built-in AI looks at the live element tree and writes a selector that targets exactly one element, even when there are dozens of buttons that look identical.

Key Benefits

- **Point and Capture** — Press F1 over any control and see its full UIA tree, properties and supported actions instantly.
- **Bulletproof Selectors with AI** — When a basic selector matches several buttons, the AI rewrites it to target exactly one — no more guessing indexes by hand.
- **Macro Builder Built In** — Add elements as steps, pick the action (Click, SetValue, Highlight, etc.), and the inspector writes the AHK code for you.
- **Anchor Points for Stable Macros** — Save a parent element as a named anchor so your scripts keep working even when surrounding controls shift.

- **Test Without Leaving the App** — Hit Test to run your macro live; the inspector highlights matched elements so you can confirm the right one fires.
- **Friendly Hover Hints** — Every control on the inspector shows a brand-colored Notify hint explaining what it does, so nothing is mysterious.

02 Installation & First Launch

2.1 System Requirements

- Windows 10 or 11
- AutoHotkey v2.0.2 or newer (only if running the `.ahk` source — the compiled `.exe` needs nothing extra)
- Optional: an [OpenRouter API key](#) for the AI features (Find Unique and Ask AI)

2.2 How to Install

1. Drop the `UIA_Inspector` folder anywhere on your machine — Desktop, Documents, a tools folder, all fine.
2. Double-click `UIA_Inspector.ahk` (or the compiled `.exe`) to launch.
3. On first run the Preferences window opens automatically — pick your settings, optionally paste an OpenRouter API key, and click Save.
4. The main inspector window appears with the title bar showing your active hotkeys (e.g. *F1 = Capture element under mouse · F2 = Toggle tracking*).

TIP — SETTINGS LIVE BESIDE THE SCRIPT

All your preferences are saved in `settings.ini` in the same folder as the inspector.
Move the folder, your settings move with it. No registry, no cloud.

03 Quick Start

The whole tool revolves around two hotkeys. Memorize these and you can do 80% of what UJA Inspector offers.

Hotkey	Action
F1	Capture the element under your mouse and load it into the inspector.
F2	Toggle live tracking — the inspector follows your mouse and updates continuously.

3.1 Tree Navigation Shortcuts

Key	What it does
Numpad +	Expand current node / jump to first child / move to next.
Numpad -	Collapse current node / jump to parent.
↑ / ↓	Move between siblings in the tree.
Ctrl + I	Jump to the nearest ancestor that has an AutomationId — your most resilient anchor.

Ctrl + Z	Undo the last macro step.
-----------------	---------------------------

TIP — THE TITLE BAR ALWAYS TELLS YOU

Whatever hotkeys you choose in Preferences are shown live in the window title and status bar — so even after you rebind F1 to something else, you'll never forget what to press.

04 Capturing Elements

4.1 The F1 Capture

Hover your mouse over any control in any application and press **F1**. The inspector populates four panels at once:

- **Window Info** — Title, class, HWND, process ID and size of the host window.
- **Properties** — Every UIA property of the captured element (Name, AutomationId, ClassName, Type, etc.).
- **Anchor Points** — Suggested parent elements that uniquely identify the area around your target.
- **Patterns** — Which actions the element supports (Click, SetValue, Toggle, Expand, etc.). Double-click any ► leaf to fire that action live.

4.2 Live Tracking with F2

Press **F2** and the inspector follows your mouse continuously — every control you hover updates the panels. Press F2 again to stop. Useful for exploring an unfamiliar app without spamming F1.

4.3 Browsers Get Special Treatment

Chromium-based browsers (Chrome, Edge, Brave) need a tiny extra step to expose UIA. The inspector detects browser windows automatically and uses the right approach so DOM elements show up just like native controls.

NOTE — ADMIN APPS NEED ADMIN INSPECTOR

If the app you're inspecting runs as Administrator, the inspector must run as Administrator too — Windows blocks UIA access otherwise. The title bar shows *(Admin)* when you're elevated.

05 The UIA Tree

The middle panel shows the entire UIA tree of the captured window. Click any node to inspect it; right-click for code generation. The captured element is highlighted so you always see where you are.

5.1 Deep Scan

By default the inspector shows controls visible in the immediate hierarchy. Tick **Deep Scan** to do a full recursive walk — slower, but reveals every descendant including off-screen items. Toggle it from either the toolbar checkbox or Preferences (where you can make it the default).

5.2 Filter Box

Type into the filter box to highlight matching nodes by name or type. Useful when a window has hundreds of controls and you're looking for one. The match count appears in the status bar.

5.3 Element Count

The little read-only field next to Deep Scan tells you how many elements were found in the current tree. Big number = complex window — consider filtering or using anchors instead of walking the whole tree at runtime.

06 Properties & Patterns

6.1 Properties Panel

Lists every UIA property of the selected element. By default only the important ones are shown (Name, AutomationId, ClassName, Type, ControlType). Tick **Show All** to see everything UIA exposes — bounding rectangle, runtime ID, accelerator key, helper text, and more.

Right-click any row to copy its value to the clipboard.

6.2 Patterns Panel

UIA "patterns" are the actions an element supports. The patterns panel lists them as a tree:

- **Invoke** — A button-like click.
- **Toggle** — On/off (checkboxes, switches).
- **SetValue** — Type into edit fields.
- **Expand / Collapse** — Tree nodes, dropdowns.
- **Selection / SelectionItem** — Lists, radio groups.
- **Scroll** — Scroll containers.

Double-click any ► leaf to fire the action immediately on the captured element — a fast way to confirm you've grabbed the right control before adding it to a

macro.

07 Anchor Points

Anchor Points are the secret to writing macros that don't break the next time the app updates. Instead of saying *"click the third button"*, you say *"find the panel called **Settings**, then click the button called **Save** inside it"*. That's an anchor.

7.1 Anchor Points Panel

After capturing, the right-hand **Anchor Points** tree lists candidate ancestors that uniquely identify a region. Click any one to inspect it; right-click to **Create Anchor** — this saves it as a named variable you can reuse in your macro.

7.2 Using Anchors in Macros

The macro builder has an **Anchor** dropdown. Pick one of your saved anchors and the search will be scoped under that element instead of the whole window — much faster, much more reliable.

7.3 Refresh / Rename / Delete

- **Refresh** — Re-scans the captured window for fresh anchor candidates.
- **Right-click an anchor in the dropdown** — Rename or delete saved anchor variables.

TIP — PREFER AUTOMATIONID ANCHORS

Anchors with an AutomationId are the most stable — they survive language changes, theme updates and minor reshuffles. Hit **Ctrl + I** to jump to the nearest ancestor that has one.

08 Building Macros

The macro toolbar runs across the middle of the window. Each control is a knob you can dial to describe how the next step should target its element.

Control	What it controls
Anchor	Which saved anchor to scope the search under. (none) = search the whole window.
Scope	Where to search. Descendants (default) = all levels. Children = direct children only.
Match	How the Name is matched. Exact (default) , Contains, StartsWith, EndsWith.
Action	What to do once found: Highlight, Click, MouseClick, ControlClick, SetValue, Invoke, Toggle, Expand, Collapse, Select, ScrollIntoView, SetFocus, WaitElement.
#	When the condition matches several elements, picks which one (1 = first).

8.1 Path Matches Counter

The little read-only field labelled **Path matches** tells you how many elements your current condition resolves to. **1** means you've nailed it. Anything higher

means the selector is ambiguous — fix it manually, bump the # field, or hit **Find Unique** for AI help.

8.2 Add Element

Hit **Add element** to push the current selection + chosen action onto the macro script. The Scintilla code editor below shows the running AHK code. Edit it directly if you want — it's just text.

8.3 Test, Copy, Clear

Button	Action
Test	Run the macro live so you can see exactly what it does.
Copy	Copy the full macro script (with <code>#Requires</code> + <code>#Include</code>) to the clipboard.
Clear	Wipe all steps and reset the editor.
Clear Inspector	Clears the captured element, properties, anchors and patterns — without touching the macro.

09 AI Assistance

The two AI buttons sit right of the Path matches counter and only light up after you've captured an element. They both call your chosen OpenRouter model — see **Preferences** → **AI** to pick the model and paste your API key.

9.1 Find Unique

This is the killer feature. When your current selector matches several elements (Path matches > 1), click **Find Unique**. The AI looks at the captured element, its window, the surrounding tree and the chain of ancestors, then writes a fresh AHK snippet that targets *exactly one* match.

The strategy it follows, in order:

1. If the element itself has a unique AutomationId — use it directly.
2. Otherwise chain from a uniquely-named ancestor.
3. As a last resort, use indexed FindAll (`FindAll (...) [N]`).

The reply is appended to the macro editor as a comment + code block, ending with `.Highlight()` so you can hit Test and visually confirm the right element.

9.2 Ask AI

Opens a persistent chat window seeded with the current element + window context. Ask follow-up questions like *"why does this fail when the panel is*

collapsed?" or "give me a SetValue version that waits 200ms first". The chat keeps history so you can iterate without re-explaining.

NOTE — API KEY LIVES IN SETTINGS.INI

Your OpenRouter API key is stored locally in `settings.ini`, never sent anywhere except OpenRouter. The Show checkbox unmask it temporarily so you can verify the paste.

10 Preferences

Open Preferences from the tray menu (right-click the tray icon → **Preferences**) or by clicking the Triggers icon. Changes are applied live — no restart needed.

10.1 Hotkeys

Rebind **Capture Element** and **Toggle Tracking** to whatever keys suit you. The window title and status bar update immediately to show the new bindings.

10.2 Application Settings

Setting	Description
Always on Top	Keep the inspector floating above other windows. (Default: on)
Deep Scan by default	Start every capture in deep-scan mode. Slower but complete. (Default: off)
Mouse Tips	Show the brand-colored Notify hints when you hover controls in the inspector. (Default: on)
Remember position	Save the inspector window's screen position between sessions. (Default: on)

Guide Mode	Highlight the status-bar hint when nothing is captured, so beginners know what to do next. (Default: on)
-------------------	---

10.3 AI (OpenRouter)

Setting	Description
Model	Pick from a curated list of OpenRouter models — Claude, GPT-4.1, Gemini, Llama, DeepSeek, Grok, Mistral. (Default: Claude Sonnet 4.5)
API Key	Your OpenRouter API key. Get one from openrouter.ai/keys . The Show checkbox un.masks it.
MaxTokens	Caps the reply length — keep lower if you have a small credit balance. (Default: 2048)



11 Tray Menu

Menu Item	Action
Preferences	Open the full settings window (hotkeys, app settings, AI).
Mouse Tips	Toggle the brand-colored Notify hints on or off without opening Preferences.
About	Show version, author and link back to the-Automator.com.

12 What's New in 2.5

- **Find Unique (AI)** — One-click AI selector refinement when your condition matches several elements.
- **Ask AI Chat** — Persistent AI chat seeded with the captured element so you can iterate naturally.
- **OpenRouter Model Picker** — Choose between Claude, GPT-4.1, Gemini, Llama, DeepSeek, Grok and Mistral right from Preferences.
- **Brand-Colored Notify Tips** — Hover hints now appear in a soft yellow Notify popup at the cursor with topic-specific headers, replacing the old plain tooltip.
- **Live Hotkey Labels** — Window title and status bar always show your current hotkeys, even after rebinding in Preferences.
- **Code-Generation Busy Indicator** — A friendly busy panel appears while AI is thinking so you know it's working, not frozen.

13 Tips & Troubleshooting

13.1 General Tips

- **Aim for Path matches = 1** — That's the gold standard for a stable selector. Anything else is a future bug waiting to happen.
- **Anchor early, anchor often** — A 3-step macro built around an AutomationId anchor will outlive a 1-step macro that walks the whole window.
- **Use Highlight first** — Before swapping in Click or SetValue, set Action to Highlight and hit Test. You'll see exactly which element will fire.
- **Hover for hints** — Every control on the inspector has a Notify hint. If you forget what something does, just point at it.
- **Press Hotkeys for the legend** — The Hotkeys button shows a floating cheat sheet of every shortcut in the inspector.

13.2 Troubleshooting

Symptom	Solution
---------	----------

F1 captures nothing or empty properties	Target app probably runs as Admin. Restart the inspector as Administrator (right-click → Run as administrator).
Browser DOM elements don't show	Make sure the browser tab is focused and try again. The inspector enables UIA on Chromium browsers automatically on capture.
Find Unique button greyed out	It only enables when Path matches > 1. If your condition already targets one element, you don't need it.
"OpenRouter error" when clicking Find Unique or Ask AI	Open Preferences and check your API key — confirm it starts with <code>sk-or-</code> and that you have credit on openrouter.ai.
Tips appear in the wrong corner	By design — the Notify popup appears next to your mouse. If you find them distracting, turn them off via tray → Mouse Tips.
Test runs but the wrong element fires	Path matches is > 1. Either bump the # field, add an anchor, or click Find Unique to let AI rewrite the selector.

*UIA Inspector is developed by **the-Automator.com***

the-Automator.com

Version 2.5.0 · CC BY 4.0